

MATH 345 Skeleton Notes

Unit 3: Beyond linear maps: local linearization

Section 3.6: Chain rule as matrix multiplication

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Unit 1 used matrix multiplication to represent composition of linear maps: applying B first and then A gives $A(B\mathbf{x}) = (AB)\mathbf{x}$. For differentiable nonlinear maps, the same idea reappears locally. The local linear maps compose, so their Jacobian matrices multiply.

The chain rule

The chain rule describes derivatives of composite functions.

Compositions

Definition: Composite Function If $G : \mathbb{R}^n \rightarrow \mathbb{R}^p$ and $F : \mathbb{R}^p \rightarrow \mathbb{R}^m$ are multivariable functions, we can form a new

 as follows. Take an input in $\mathbb{R}^n$, and then apply G to the input to obtain a vector in $\mathbb{R}^p$. Apply F to this vector to obtain a vector in $\mathbb{R}^m$. This is the output of the composite function. In other words, our input is fed through the following diagram

$$\mathbb{R}^n \xrightarrow{G} \mathbb{R}^p \xrightarrow{F} \mathbb{R}^m.$$

We write the composite function as $F \circ G$ (the symbol $\circ$ should be read as “composed with”). In symbols, the composite function is defined as

$$(F \circ G)(\mathbf{x}) = F(G(\mathbf{x})).$$

Note Note that the composite $F \circ G$ is *only* defined when the output of G is a possible input to F , i.e., the output of G is $\mathbb{R}^p$, which is the space of inputs to F .

Unit 1 used matrix multiplication to represent composition of linear maps: applying B first and then A gives $A(B\mathbf{x}) = (AB)\mathbf{x}$.

Activity

Let $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and $T_2 : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be linear maps defined by their component functions as follows:

$$T_1(x, y) = \begin{bmatrix} 2x - y \\ 3y \\ x + y \end{bmatrix} \quad T_2(u, v, w) = \begin{bmatrix} u + v - w \\ 2u - v \end{bmatrix}.$$

Find the matrices A_1 and A_2 associated with these maps, so that $T_1 = T_{A_1}$ and $T_2 = T_{A_2}$ (recall the notation from Matrix map).

Activity

Let $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and $T_2 : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be linear maps defined by their component functions as follows:

$$T_1(x, y) = \begin{bmatrix} 2x - y \\ 3y \\ x + y \end{bmatrix} \quad T_2(u, v, w) = \begin{bmatrix} u + v - w \\ 2u - v \end{bmatrix}.$$

Calculate the composite map $T_2 \circ T_1$ and express it as a matrix map from $\mathbb{R}^2$ to $\mathbb{R}^2$.

Here is a nonlinear example of the composition of two functions.

Activity

Consider $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $G : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$F(u, v) = (uv, u + v) \quad \text{and} \quad G(x, y) = (e^{xy}, x - y).$$

Find expressions for $(F \circ G)(x, y)$ and $(G \circ F)(u, v)$.

Warning Observe that $F \circ G$ and $G \circ F$ are very different functions. We must therefore be careful that we compose functions in the correct order. The order matters. The rightmost Jacobian matrix acts first on the input change.

Activity

Consider $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^3$ and $\mathbf{p} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by

$$\mathbf{r}(t) = (t, \cos(t), \sin(t)) \quad \text{and} \quad \mathbf{p}(x, y, z) = (y, z)$$

Find an expression for $(\mathbf{p} \circ \mathbf{r})(t)$.

Activity

Consider $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^3$ and $\mathbf{p} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by

$$\mathbf{r}(t) = (t, \cos(t), \sin(t)) \quad \text{and} \quad \mathbf{p}(x, y, z) = (y, z)$$

Is the composition $\mathbf{r} \circ \mathbf{p}$ defined? Why or why not?

Chain rule for one-parameter inputs

Recall the chain rule from single-variable calculus, that for two functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$, we have

$$(f \circ g)'(t) = (f' \circ g)(t)g'(t).$$

The next theorem is the component form of the chain rule.

Recall that for a function $f(x, y)$, $D_1 f = f_x = \frac{\partial f}{\partial x}$, and that $D_2 f = f_y = \frac{\partial f}{\partial y}$.

Theorem: Chain Rule for One Independent Variable Consider a scalar-valued function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, and a vector-valued function $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^2$ of a single variable. Then the composition $f \circ \mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}$ is a scalar-valued function of a single variable. If $\mathbf{r}$ is differentiable at t_0 , and f is differentiable at $\mathbf{r}(t_0)$, then $f \circ \mathbf{r}$ is differentiable at t_0 and

$$(f \circ \mathbf{r})'(t_0) = (D_1 f \circ \mathbf{r})(t_0)r_1'(t_0) + (D_2 f \circ \mathbf{r})(t_0)r_2'(t_0),$$

where r_1 and r_2 are the components of $\mathbf{r}$.

Proof We note that the right hand side of the equation can be written as $L \circ \mathbf{r} - f(\mathbf{r}(t_0))$, where L is the linear approximation of f at $\mathbf{r}(t_0)$. Because f is differentiable, we know that

$$\lim_{\mathbf{x} \rightarrow \mathbf{r}(t_0)} \frac{f(\mathbf{x}) - L(\mathbf{x})}{\|\mathbf{x} - \mathbf{r}(t_0)\|} = 0.$$

Now

$$\lim_{t \rightarrow t_0} \frac{\|\mathbf{r}(t) - \mathbf{r}(t_0)\|}{t - t_0} = \left\| \lim_{t \rightarrow t_0} \frac{\mathbf{r}(t) - \mathbf{r}(t_0)}{t - t_0} \right\| = \|\mathbf{r}'(t_0)\|.$$

Using these two equations (substituting $\mathbf{r}(t)$ for $\mathbf{x}$ in the first equation), we find that

$$\begin{aligned} \lim_{t \rightarrow t_0} \frac{f(\mathbf{r}(t)) - L(\mathbf{r}(t))}{t - t_0} &= \lim_{t \rightarrow t_0} \frac{f(\mathbf{r}(t)) - L(\mathbf{r}(t))}{\|\mathbf{r}(t) - \mathbf{r}(t_0)\|} \frac{\|\mathbf{r}(t) - \mathbf{r}(t_0)\|}{t - t_0} \\ &= 0 \cdot \|\mathbf{r}'(t_0)\| = 0. \end{aligned}$$

But recalling that

$$L(x, y) = f(\mathbf{r}(t_0)) + D_1 f(\mathbf{r}(t_0))(x - r_1(t_0)) + D_2 f(\mathbf{r}(t_0))(y - r_2(t_0)),$$

we conclude that

$$\begin{aligned} \lim_{t \rightarrow t_0} \frac{f(\mathbf{r}(t)) - f(\mathbf{r}(t_0))}{t - t_0} &= \lim_{t \rightarrow t_0} \frac{f(\mathbf{r}(t)) - f(\mathbf{r}(t_0))}{t - t_0} - \frac{f(\mathbf{r}(t)) - L(\mathbf{r}(t))}{t - t_0} \\ &= \lim_{t \rightarrow t_0} \frac{L(\mathbf{r}(t)) - f(\mathbf{r}(t_0))}{t - t_0} \\ &= \lim_{t \rightarrow t_0} D_1 f(\mathbf{r}(t_0)) \frac{r_1(t) - r_1(t_0)}{t - t_0} \\ &\quad + D_2 f(\mathbf{r}(t_0)) \frac{r_2(t) - r_2(t_0)}{t - t_0} \\ &= f_x(\mathbf{r}(t_0))r'_1(t_0) + D_2 f(\mathbf{r}(t_0))r'_2(t_0). \end{aligned}$$

Note There are several ways to aid in remembering the formula for the chain rule. The first is to simplify the equation, writing it as

$$(f \circ \mathbf{r})'(t) = (\nabla f \circ \mathbf{r})(t) \cdot \mathbf{r}'(t),$$

so that the equation looks more like the chain rule from single variable calculus. The second is the following mnemonic: if we write z for $f \circ \mathbf{r}$ and for f , and write the components of $\mathbf{r}$ as x and y , then the chain rule reads that

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt},$$

so that the symbols dx and dy in the numerators on the right hand side, and the symbols ∂x and ∂y that occur in the denominators appear to ‘cancel’ like when multiplying fractions.

Activity

Suppose $z = x^2y - y^2x$, where $x = \sin t$ and $y = e^t$. Compute dz/dt .

Activity

If $f(x, y) = x^2 + y^2$, and $\mathbf{r}(t) = (\sin t, \cos(2t))$, find $(f \circ \mathbf{r})'(t)$.

Chain rule for two independent variables

Theorem: Chain Rule for Two Independent Variables Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a scalar-valued function, and $G : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a vector-valued function. Then $f \circ G : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a scalar-valued function, and the two equations

$$D_1(f \circ G) = (D_1 f \circ G)D_1 G_1 + (D_2 f \circ G)D_1 G_2$$

and

$$D_2(f \circ G) = (D_1 f \circ G)D_2 G_1 + (D_2 f \circ G)D_2 G_2$$

hold.

Proof Fix the second variable, and apply Chain Rule for One Independent Variable to the first to obtain the formula for $(f \circ G)_x$. Then fix the first variable and apply Chain Rule for One Independent Variable to the second variable to obtain the formula for $(f \circ G)_y$.

Remark Recall from Section 3.4 that the Jacobian matrix collects the partial derivatives of a vector-valued map. For G , this gives

$$J_G = \begin{bmatrix} D_1 G_1 & D_2 G_1 \\ D_1 G_2 & D_2 G_2 \end{bmatrix}.$$

Then we can write the pair of equations in Chain Rule for Two Independent Variables as the single vector equation

$$\nabla(f \circ G) = J_G^T(\nabla f \circ G),$$

where we interpret ∇f as a column vector.

The second method is mnemonic. If we let x and y be the variables of f , and u and v be the variables of G , and we write z for $f \circ G$ and for f , and write the components of $G(u, v)$ as x and y , then the chain rule reads that

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u}$$

and that

$$\frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v},$$

so that the partials in the numerator and denominator again appear to ‘cancel’.

Activity

Suppose that

$$z = x^2 + xy - y^2$$

that $x = e^{2s+t}$, and that $y = \ln(t/s)$. Compute

$$\frac{\partial z}{\partial s} \quad \text{and} \quad \frac{\partial z}{\partial t}.$$

First compute the partial derivatives of z in x and y , and the partial derivatives of x and y in t and s .

Activity

Suppose that

$$z = x^2 + xy - y^2$$

that $x = e^{2s+t}$, and that $y = \ln(t/s)$. Compute

$$\frac{\partial z}{\partial s} \quad \text{and} \quad \frac{\partial z}{\partial t}.$$

Now use the chain rule to compute the partial derivatives of z in s and t .

Activity

Use the Jacobian matrix version of the chain rule to calculate the partial derivatives of the function $f \circ G$, where $f(x, y) = 3x^2 - 2xy + y^2$, and $G(u, v) = (3u + 2v, 4u - v)$.

Generalized chain rule

Before multiplying Jacobian matrices, check the input and output dimensions of each map.

Theorem: Generalized Chain Rule Suppose $F : \mathbb{R}^p \rightarrow \mathbb{R}^m$ and $G : \mathbb{R}^n \rightarrow \mathbb{R}^p$ are differentiable functions. Then $F \circ G : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differ-

entiable, and

$$J_{F \circ G}(\mathbf{x}) = J_F(G(\mathbf{x}))J_G(\mathbf{x}),$$

where the right hand side is defined by matrix multiplications. Equivalently, for $1 \leq i \leq m$,

$$D_j\{F_i \circ G\}(\mathbf{x}) = \sum_{k=1}^p (D_k F_i)(G(\mathbf{x}))(D_j G_k)(\mathbf{x}).$$

The order matters. The rightmost Jacobian matrix acts first on the input change.

Note If f is a scalar-valued function, then the generalized chain rule tells us that

$$\nabla(f \circ G) = J_G^T(\nabla f \circ G)$$

as in the two variable case.

Activity

Suppose $f(x, y, z) = 3x^2 - 2xy + 4z^2$, and that $G(u, v) = (e^u \sin v, e^u \cos v, e^u)$. Compute the partial derivatives of $f \circ G$.

Note The generalized chain rule says that local linear maps compose. The matrix formulation records that composition by multiplying Jacobian matrices in the correct order.

A tiny sigmoid block

A small neural-network block can be read as a composition of three steps: an affine map, a coordinatewise nonlinear rule, and another affine map.

Note: Sigmoid and its derivative Let

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$

The derivative is

$$\sigma'(t) = \sigma(t)(1 - \sigma(t)).$$

When σ is applied to a vector, it is applied coordinate by coordinate.

Activity: A tiny sigmoid block

Let

$$W_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad W_2 = \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 1/2 \end{bmatrix},$$

and let $\mathbf{b}_1 = \mathbf{0}$ and $\mathbf{b}_2 = \mathbf{0}$.

Define

$$\mathbf{s} = W_1 \mathbf{x} + \mathbf{b}_1, \quad \mathbf{u} = \sigma(\mathbf{s}), \quad N(\mathbf{x}) = W_2 \mathbf{u} + \mathbf{b}_2,$$

where σ is applied coordinatewise.

At the base point

$$\mathbf{a} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

do the following.

1. Compute the pre-activation vector $\mathbf{s}_a = W_1 \mathbf{a} + \mathbf{b}_1$.
2. Compute the hidden vector $\mathbf{u}_a = \sigma(\mathbf{s}_a)$.
3. Compute $N(\mathbf{a})$.
4. Use $\sigma'(0) = 1/4$ to compute

$$D_a = \text{diag}(\sigma'((\mathbf{s}_a)_1), \sigma'((\mathbf{s}_a)_2), \sigma'((\mathbf{s}_a)_3)).$$

5. Use the chain rule to compute

$$J_N(\mathbf{a}) = W_2 D_a W_1.$$

6. Use

$$N(\mathbf{a} + \mathbf{h}) \approx N(\mathbf{a}) + J_N(\mathbf{a})\mathbf{h}$$

to predict the output change¹² caused by a small input change $\mathbf{h}$.



Note: Why sigmoid here The sigmoid function is smooth, so the Jacobian matrix of this block is defined at every input. Some neural-network models use nonsmooth activation rules. Those require extra case analysis. Here the point is the chain rule for Jacobian matrices.

Activity: A one-query softmax weighting map

A single-query attention-style map can be viewed as a nonlinear vector-valued map.

Let

$$K = \begin{bmatrix} 3 & 0 \\ 0 & 3 \\ -3 & -3 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

For a query vector

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix},$$

define the score vector

$$\mathbf{s} = K\mathbf{q},$$

the softmax weight vector

$$\alpha(\mathbf{q}) = \text{softmax}(\mathbf{s}),$$

and the output

$$F(\mathbf{q}) = V^T \alpha(\mathbf{q}).$$

1. Compute $\mathbf{s} = K\mathbf{q}$ in terms of q_1 and q_2 .
2. What are the rows of V ?
3. Explain why $F(\mathbf{q})$ is a weighted average of the rows of V .
4. Compute $\alpha(\mathbf{0})$ and $F(\mathbf{0})$.
5. Explain why F is not a linear map.
6. What operation makes F nonlinear?

Tags. [U3-L05, U3-L07, U1-L08 | C+R | Core]

Note: Softmax outputs stay in the value triangle The output always lies in the triangle spanned by the three value vectors. The query vector does not choose fixed weights. It changes the scores, and the softmax rule changes all weights together.

Note: An attention-style nonlinear block Unit 1 treated attention as three visible operations:

dot-product scores $\longrightarrow$ weights $\longrightarrow$ weighted averages.

For a sequence of token vectors stored as rows of a matrix X , fixed weight matrices produce

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

The score matrix is

$$S = QK^T.$$

The entry S_{ij} compares the query for token i with the key for token j . A weighting rule, such as row-wise softmax, converts each row of S into weights. Write the resulting attention-weight matrix as A_{att} . The attention output is

$$H = A_{\text{att}}V.$$

If A_{att} were fixed, then $V \mapsto A_{\text{att}}V$ would be a linear map: each row of H would be a weighted average of rows of V . In attention, A_{att} is not fixed. It is computed from X through Q , K , scores, and softmax. Thus the full rule $X \mapsto H$ is generally nonlinear.

The Unit 3 question is local. Near one fixed input X_0 , a nonlinear rule can be approximated by a linear map. After the entries of X and H are arranged as vectors, that local linear map is represented by a Jacobian matrix.

Activity: What depends on the input?

Assume the weight matrices W_Q , W_K , and W_V are fixed. The input is the token matrix X . Define

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V,$$

$$S = QK^T, \quad A_{\text{att}} = \text{softmax}_{\text{rows}}(S), \quad H = A_{\text{att}}V.$$

1. Which objects are fixed before the input X is chosen?
2. Which objects are computed from X ?
3. Why is $X \mapsto XW_Q$ a linear map when W_Q is fixed?
4. Why does $S = QK^T$ depend on X in two places?
5. If A_{att} were fixed, what would $H = A_{\text{att}}V$ do to the rows of V ?
6. In actual attention, why is the full rule $X \mapsto H$ not just one linear map?
7. What Unit 3 object describes the local linear approximation near a fixed input X_0 ?

Tags. [U3-L05, U3-L06, U3-L07 | C+R | Core]



Warning: Specify the input Before asking whether a rule is linear, first say what is variable and what is fixed. The rule $V \mapsto A_{\text{att}} V$ is linear when A_{att} is fixed. The attention rule $X \mapsto A_{\text{att}}(X)V(X)$ is generally nonlinear because the weights and values both depend on X .

Activity: Comparing square-grid visualizations

This unit used three square-grid visualizations: the nonlinear shear from Corners are not enough, the sigmoid block from A tiny sigmoid block, and the attention-style softmax map from A one-query softmax weighting map.

Map A:

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

Map B: a two-output sigmoid block

$$\mathbf{s} = W_1 \mathbf{x} + \mathbf{b}_1, \quad \mathbf{u} = \sigma(\mathbf{s}), \quad N(\mathbf{x}) = W_2 \mathbf{u} + \mathbf{b}_2.$$

Map C: an attention-style softmax map

$$F_{\text{att}}(\mathbf{q}) = V^T \text{softmax}(K\mathbf{q}).$$

1. Which map agrees with a linear shear on the four corners but not at an edge midpoint?
2. Which map is built from affine maps with a coordinatewise sigmoid between them?
3. For which map is the local matrix computed by $W_2 D_a W_1$?
4. Which map produces outputs inside the triangle spanned by value vectors?
5. For which map does the nonlinear step come from softmax weights that depend on the input?
6. In all three examples, what does the square-grid visualization show that four corners alone may miss?

Tags. [U3-L01, U3-L05, U3-L07, U1-L08 | C+R | Core]

